Two connected countermodels to one-circuit repair for five-cycle double covers

AI-assisted research draft for independent human verification

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AI-assistance disclosure. OpenAI Codex agents (GPT-5-series models, accessed 25–26 July 2026) proposed the successful-value and one-circuit-repair formulations, formulated the pure-coordinate-merge strategy, generated proof ideas and substantial prose, wrote the producer and checking programs, performed the computational searches and literature audit, found the finite countermodels, designed the two-sum constructions, discovered the 40- and 46-vertex reductions and the rigid-cap obstruction, and generated the checking artifacts, all under human direction. A second Codex agent was asked to audit the logic adversarially. The "independent" Python checker, the separately written C++ implementation, and that audit were also AI-assisted; none is independent human verification or peer review. The main results refute two intermediate one-circuit strategies, not the five-cycle double cover conjecture; both constructed graphs positively have standard five-cycle double covers. No human author or referee is represented as having certified the manuscript, proof, certificate, or computation. This research draft is not journal-ready and should not be submitted until human authors check every proof and incidence table, independently reconstruct the graph, rerun the computations, verify the literature attribution, and accept ordinary scholarly responsibility.

Abstract

We give two explicit countermodels to one-circuit repair strategies for the five-cycle double cover conjecture. The first uses a nowhere-zero $\mathbb{F}_2^3$-flow f on a simple connected bridgeless cubic 40-vertex graph. No value class of f, or of any allowed neighbour obtained by switching one connected circuit, satisfies a particular edge-disjoint T-join packing condition sufficient for a 5-CDC.

The second concerns the potential construction of an eight-cover from a fixed $\mathbb{F}_2^3$-flow used in recent proofs and expositions of the cycle double cover theorem. On an explicit simple connected bridgeless cubic 46-vertex graph, no compatible eight-cover for the displayed flow can be purely merged into at most five Eulerian subgraphs. The same remains true after every nowhere-zero switch on one connected circuit. A 12-vertex port-deleted block is potential-rigid up to coordinate translation and forces a K_6 in the coordinate co-occurrence graph. Three copies are attached at three same-colour base edges that no circuit can contain simultaneously; a short triangle argument proves this non-cyclability. These ingredients give SAT-free proofs of both global one-switch assertions.

Both countermodels are Tait-colourable and have explicit three-cycle double covers. They therefore refute only the stated intermediate strategies and do not resolve or refute the five-cycle double cover conjecture.

For context, exact enumeration had found no obstruction through all connected simple bridgeless cubic graphs of order at most 14, selected complete hard-graph scopes through order 20, and all 20 strict snarks of order 22 and the 38 strict order-24 snarks output by a retained

Snarkhunter 2.0b run. Connected-circuit enumeration checks 847,539,220 flow-orbit representatives at order 20, 63,251,330 at order 22, and 617,079,260 at order 24. The 40-vertex construction shows that these are finite phenomena, not a universal theorem. All artifact generation, checking code, and proof development were AI-assisted and still require independent human verification.

1 Introduction and attribution

A *standard five-cycle double cover* (5-CDC) of a finite graph is a list $C_1, \dots, C_5$ of Eulerian edge-subsets such that every edge belongs to exactly two members. Here Eulerian means even degree at every vertex; connectedness is not required. Empty members are permitted, so a cover with at most five members can be padded to five. This note concerns the standard, unoriented conjecture only. This agrees with the current "at most k" convention in Oum's exposition [9, Section 9.1].

The existence of a 5-CDC for every bridgeless graph remains unproved. Oum's July 2026 exposition gives an 8-CDC and records the 5-CDC statement separately as Conjecture 18 [9, Section 9.1]. See also the contemporary expositions of Geelen and Kintali [3, 6]. The present note does not claim otherwise. Its two main results instead refute two specific intermediate proof strategies inside the space of nowhere-zero $\mathbb{F}_2^3$-flows. Both countermodels have explicitly checked 5-CDCs.

The underlying matching/nowhere-zero-4-flow characterization is prior: Hoffmann-Ostenhof proves that a cubic graph has a 5-CDC precisely when it has a matching M, two Eulerian edge-subsets intersecting exactly in M, and a nowhere-zero 4-flow on $G-M$ [4, Corollary 0.6]. We restate and prove the exact specialization needed below to eliminate convention ambiguity. Further 4-flow approaches and sufficient conditions for 5-CDCs, including Catlin-reduction and weak-oddness results, are due to Liu, Hao, Luo, and Zhang [7, 8]. None of those results is claimed as new here.

The reconfiguration language is also prior. Esperet, Hendrey, Lagoutte, Marseloo, Norin, and Steiner define the graph of nowhere-zero flows in which consecutive flows differ on one circuit and prove positive and negative connectivity results [2]. Cranston, Li, Su, Wang, and Xu develop this theory further, including reductions to cubic graphs and results for several groups and graph classes [1]. In particular, the definition of circuit-supported adjacency is not a contribution of this note. More precisely, Esperet et al. prove that for every 2-edge-connected graph the reconfiguration graph has minimum degree at least one when the abelian flow group has order at least six or is $\mathbb{Z}_2^2$; they also treat $\mathbb{Z}_4$, while frozen $\mathbb{Z}_5$-flows exist [2]. Even the positive minimum-degree statements are much weaker than requiring every unsuccessful flow to have a *successful* neighbour. Their universal vector-space connectivity result uses $\mathbb{Z}_2^8$, not $\mathbb{F}_2^3$.

The term *Fano-flow conjectures* also has an established, different meaning: Jin, Mazzuoccolo, and Steffen study conjectures controlling how many lines of the Fano plane are used by a flow [5]. Our successful-value condition is instead a value-class/T-join packing condition, and (D) is a domination statement in a reconfiguration graph. Neither countermodel below refutes any of those classical Fano-flow conjectures.

We found no statement in the sources above of either specific one-circuit domination assertion, the pure-merge coloring obstruction, or the two countermodels proved here. Accordingly, the only priority claim made here is provisional: these two negative results appear not to have been recorded. The successful-value definition, exact blocks, two-sum compositions, tetrahedral T-join observation, and finite computations belong to those constructions; the underlying flow, T-join, two-sum, potential, and circuit-reconfiguration machinery is not claimed as new. In particular, the flow-to-pair-label potential construction appears in Oum's and Kintali's presentations [9, Section 5] and [6]. An independent expert literature review, including consultation with authors of the recent flow and reconfiguration papers, is required before submission.

2 Pair labels and the exact Boolean encoding

We first fix the direct 5-CDC convention independently of the later flow reformulation. For this section an undirected loop contributes its two incidences at its vertex. The remaining sections and all computations are loopless.

Proposition 2.1 (pair-label equivalence)**.** *For a finite undirected graph G, the following are equivalent.*

1. *G has Eulerian edge-subsets $C_1, \dots, C_5$ and every edge belongs to exactly two of them.*
2. *Every edge e has a label $L(e) \in \binom{[5]}{2}$, and for every vertex v and coordinate $i \in [5]$, coordinate i occurs evenly among the labels on the edge incidences at v.*

Proof. Given the cover, put $L(e) = \{i : e \in C_i\}$. Exact double coverage gives $|L(e)| = 2$, and the Eulerian condition on C_i gives the required incidence parity. Conversely, put $C_i = \{e : i \in L(e)\}$. The incidence parity makes every C_i Eulerian, and $|L(e)| = 2$ makes every edge belong to exactly two members. □

This equivalence has a literal SAT/XOR encoding. Introduce $x_{e,i} \in \{0, 1\}$, with $x_{e,i} = 1$ meaning $i \in L(e)$, and impose

$$\sum_{i=1}^{5} x_{e,i} = 2 \quad (e \in E(G)), \qquad \bigoplus_{e \ni v} x_{e,i} = 0 \quad (v \in V(G),\ i \in [5]). \tag{1}$$

Incidences, rather than merely edge names, are counted in the second equation. Proposition 2.1 proves that (1) is equivalent to the intended definition.

For completeness, the exactly-two condition on five variables has the certificate-producing CNF

$$\neg x_i \vee \neg x_j \vee \neg x_k \quad (1 \le i < j < k \le 5), \tag{2}$$

together with

$$\bigvee_{j \ne i} x_j \quad (i \in [5]). \tag{3}$$

The ten clauses (2) give at most two true variables. If zero variables are true, every clause (3) fails; if only x_i is true, the clause omitting i fails. Hence (2)–(3) give exactly two.

At a cubic vertex, writing one coordinate on its incident edges as a, b, c, even parity is exactly the four-clause CNF

$$(a \vee b \vee \neg c), \quad (a \vee \neg b \vee c), \quad (\neg a \vee b \vee c), \quad (\neg a \vee \neg b \vee \neg c). \tag{4}$$

These clauses exclude the four odd assignments. Equations (1), or their CNF expansion (2)–(4), are the authoritative standard-cover semantics; no orientation variables occur.

3 Value classes and T-joins

From now on G is a finite loopless cubic graph and

$$f : E(G) \longrightarrow A - \{0\}, \qquad A = \mathbb{F}_2^3,$$

is a nowhere-zero flow. Since A has exponent two, edge orientations play no role: at every vertex the xor of the three incident values is zero. For $k \in A - \{0\}$, put

$$M_k = \{e : f(e) = k\}.$$

For an edge set X, write ∂X for the set of vertices incident with an odd number of edges of X. If $T \subseteq V(H)$, a *T-join* of H is an edge set J with $\partial J = T$.

Proposition 3.1 (value-class structure)**.** *For every $k \neq 0$:*

1. *M_k is a matching;*
2. *the quotient of f by $\langle k \rangle$ is an $\mathbb{F}_2^2$-flow whose exact zero set is M_k;*
3. *$G - M_k$ has a canonical family of four ∂M_k-joins, indexed by the linear functionals*

$$\Lambda_k = \{\lambda \in A^* : \lambda(k) = 1\};$$

4. *these four joins double-cover $G - M_k$, and each unordered pair of joins intersects in one of the other six value classes.*

Proof. At a cubic vertex the three incident nonzero values sum to zero. Two cannot be equal, since the third would then be zero. Thus a fixed value occurs at most once at a vertex, proving that M_k is a matching.

The zero coset of $A/\langle k \rangle$ is $\{0, k\}$. Since f is nowhere zero, an edge maps to zero in the quotient exactly when its original value is k. This proves the second assertion.

There are four linear functionals in Λ_k. For $\lambda \in \Lambda_k$, the support of the binary flow $\lambda \circ f$ is an Eulerian edge set containing M_k. Therefore

$$J_\lambda = \{e \notin M_k : \lambda(f(e)) = 1\}$$

has boundary ∂M_k in $G - M_k$.

If $f(e) = w \notin \{0, k\}$, the two independent equations $\lambda(k) = \lambda(w) = 1$ have exactly two solutions in the three-dimensional dual space. Hence e occurs in exactly two of the four joins. Finally, if $\lambda \neq \mu$ belong to Λ_k, the simultaneous equations

$$\lambda(x) = \mu(x) = 1$$

have exactly two solutions in A: k and one value $w \notin \{0, k\}$. After M_k is deleted, $J_\lambda \cap J_\mu = M_w$. The six unordered pairs of functionals give the six values $w \neq 0, k$. □

In particular, the elementary component-parity obstruction to the existence of even one T-join never occurs for a value class. The issue is whether two such joins can be made edge-disjoint; the four canonical joins need not contain a disjoint pair.

Lemma 3.2 (even-marked circuit criterion)**.** *Let M be a matching in a cubic graph G, put $K = G - M$ and $T = \partial M$. Then K has two edge-disjoint T-joins if and only if K has an Eulerian edge set Q that contains every vertex of T and whose every nonempty circuit component contains an even number of vertices of T.*

Proof. Suppose J_1, J_2 are edge-disjoint T-joins. A terminal has degree two in K; each join has odd degree there, so each uses one edge, and disjointness makes them use the two different edges. At a nonterminal, each join has even degree in a three-edge incidence set. Thus $Q = J_1 \dot\cup J_2$ has degree two at every terminal and degree zero or two elsewhere. It is a disjoint union of circuits. Along a circuit, the join colour stays unchanged at a nonterminal and switches at a terminal. The colour can close consistently only after an even number of switches, so every circuit contains evenly many terminals.

Conversely, traverse each circuit of Q, colouring its edges with two join colours. Keep the colour through a nonterminal and switch it at a terminal. The even terminal count makes the colouring consistent when the traversal closes. Each colour class has degree one at every terminal and degree zero or two elsewhere, so the two colour classes are edge-disjoint T-joins. ☐

Definition 3.3. A value $k \neq 0$ is *successful* for f if $(G - M_k, \partial M_k)$ has two edge-disjoint T-joins. The flow f is *good* if it has at least one successful value.

Proposition 3.4 (successful value implies a 5-CDC)**.** *If f is good, then G has a standard five-cycle double cover.*

Proof. Let k be successful, put $M = M_k$, and take edge-disjoint T-joins J_1, J_2 in $G - M$, where $T = \partial M$. Then

$$A_1 = M \dot\cup J_1, \qquad A_2 = M \dot\cup J_2$$

are Eulerian edge sets of G, and $A_1 \cap A_2 = M$.

Let $\phi : E(G) \to \mathbb{F}_2^2$ be the quotient flow from Proposition 3.1; its exact zero set is M. On $K = G - M$, put

$$D_0 = A_1 \triangle A_2 = J_1 \dot\cup J_2.$$

We construct a four-cycle double cover of K whose first coordinate is D_0. Represent the nonzero elements $1, 2, 3$ of $\mathbb{F}_2^2$ by the linear section

$$s(1) = 1100, \qquad s(2) = 1010, \qquad s(3) = 0110$$

into the even-weight subspace of $\mathbb{F}_2^4$, and put $r = 1111$. Let $\ell(y)$ be the first coordinate of $s(y)$. Define

$$h(e) = 1_{D_0}(e) + \ell(\phi(e)), \qquad d(e) = s(\phi(e)) + h(e)r. \tag{5}$$

Both D_0 and the support of the binary flow $\ell \circ \phi$ are Eulerian, so h is a binary flow. Hence d is an $\mathbb{F}_2^4$-flow. For $\phi(e) \neq 0$, the two vectors $s(\phi(e))$ and $s(\phi(e)) + r$ are complementary weight-two vectors. Thus every $d(e)$ has weight two. Its first coordinate is

$$\ell(\phi(e)) + h(e) = 1_{D_0}(e).$$

The four coordinate supports D_0, D_1, D_2, D_3 of d therefore form a four-cycle double cover of K with prescribed member D_0.

Now take

$$A_1,\ A_2,\ D_1,\ D_2,\ D_3.$$

An edge of M lies in A_1, A_2 and nowhere else. An edge of D_0 lies in exactly one of A_1, A_2 and exactly one of D_1, D_2, D_3. An edge outside $D_0 \cup M$ lies in neither A_1, A_2 and in exactly two of D_1, D_2, D_3. Every edge is covered exactly twice. ☐

The conclusion and its matching/4-flow ingredients are consistent with, and are a direct specialization of, the prior characterization in [4]. The detailed proof is included only to make the coordinate convention and the role of the two T-joins checkable.

4 Circuit switches and the false domination statement

Let $a \neq 0$, and let C be a circuit in G. Define

$$f^{C,a}(e) = \begin{cases} f(e) + a, & e \in C, \\ f(e), & e \notin C. \end{cases} \tag{6}$$

Lemma 4.1 (exact one-circuit exchange)**.** *If $C \cap M_a = \varnothing$, then $f^{C,a}$ is a nowhere-zero flow, $M'_a = M_a$, and, for $b \notin \{0, a\}$,*

$$M'_b = (M_b - C) \cup (M_{a+b} \cap C). \tag{7}$$

Proof. At every vertex, the circuit has either zero or two incident edges. Adding a on those two edges changes the conservation sum by $a + a = 0$, so (6) remains a flow. A switched edge becomes zero exactly when it previously had value a, excluded by the hypothesis.

No edge of M_a is switched. No switched nonzero value can become a, since $x + a = a$ would imply $x = 0$. Thus $M'_a = M_a$. For $b \notin \{0, a\}$, an edge outside C retains value b, while an edge of C has new value b exactly when its old value was $a + b$. This is (7). □

The recent reconfiguration papers [2, 1] write $\mathcal{F}(G, A)$ for the graph whose vertices are nowhere-zero A-flows and whose adjacent flows differ only on a circuit. For $A = \mathbb{F}_2^3$, Lemma 4.1 describes exactly those adjacencies. Indeed, if the difference of two A-flows is supported on a connected circuit, conservation at a circuit vertex equates the difference values on its two circuit edges; connectedness makes the difference constant around the circuit.

Let $\mathcal{G}(G)$ be the set of good nowhere-zero $\mathbb{F}_2^3$-flows.

One-circuit domination statement (D)**.** *For every finite connected bridgeless loopless cubic graph G, the set $\mathcal{G}(G)$ dominates $\mathcal{F}(G, \mathbb{F}_2^3)$.*

Thus every nowhere-zero flow would either be good or have a good neighbour obtained by one allowed connected-circuit switch. This statement is weaker than connectivity of the reconfiguration graph and is not implied by the known absence of isolated flows. More importantly, Theorem 6.4 below proves that (D) is false even for simple graphs.

Proposition 4.2 (why (D) would have sufficed)**.** *If* (D) *were true, then every finite connected bridgeless loopless cubic graph has a standard 5-CDC.*

Proof. Every bridgeless graph has a nowhere-zero 6-flow by Seymour's theorem [10], hence a nowhere-zero 8-flow. Tutte's group-order invariance for nowhere-zero flows [11] supplies a nowhere-zero $\mathbb{F}_2^3$-flow. If it is good, apply Proposition 3.4. If it is bad, (D) supplies a good adjacent flow in one move, and the same proposition applies. □

Theorem 6.4 closes this particular route negatively. It says nothing negative about the conclusion of Proposition 4.2; indeed, its countermodel graph has a standard 5-CDC.

5 A human-checkable bad supplied flow

This section disproves the stronger assertion that every supplied nowhere-zero $\mathbb{F}_2^3$-flow is already good. The proof is finite and does not depend on SAT or on the frontier program.

Theorem 5.1 (countermodel to the every-flow claim)**.** *There is a connected simple bridgeless cubic graph G with a nowhere-zero $\mathbb{F}_2^3$-flow f for which none of the seven nonzero values is successful. The same graph is 3-edge-colourable and has an explicit standard 5-CDC.*

Proof. Number the edges and assign values as follows. The final column will be used for a positive-control 3-edge-colouring.

i	e_i	$f(e_i)$	$c(e_i)$	i	e_i	$f(e_i)$	$c(e_i)$
0	03	7	0	8	28	3	2
1	06	6	1	9	36	5	2
2	07	1	2	10	39	2	1
3	14	7	2	11	48	1	1
4	16	3	0	12	49	6	0
5	17	4	1	13	58	2	0
6	25	6	1	14	59	4	2
7	27	5	0				

(8)

Every vertex occurs in three rows, so G is cubic and simple. The vertex sequence

$$0, 3, 6, 1, 4, 9, 5, 8, 2, 7, 0$$

is a Hamilton circuit. Its edges lie on that circuit, and every remaining edge is a chord whose union with either Hamilton path between its ends contains a circuit. Hence every edge lies on a circuit and G is bridgeless. Its graph6 encoding is `IC0ef?kF?`.

At vertices $0, \ldots, 9$, the incident value triples are

$$\begin{gathered}(7,6,1),\ (7,3,4),\ (6,5,3),\ (7,5,2),\ (7,1,6),\\ (6,2,4),\ (6,3,5),\ (1,4,5),\ (3,1,2),\ (2,6,4).\end{gathered} \tag{9}$$

Each triple has xor zero and contains no zero, proving directly that f is a nowhere-zero flow. Its value classes are

k	M_k	k	M_k
1	$\{2, 11\}$	5	$\{7, 9\}$
2	$\{10, 13\}$	6	$\{1, 6, 12\}$
3	$\{4, 8\}$	7	$\{0, 3\}$
4	$\{5, 14\}$		

(10)

and the endpoint pairs in every displayed class are disjoint.

It remains to prove that every class fails the criterion of Lemma 3.2. Fix k, put a binary variable q_i on edge i, set $q_i = 0$ for $i \in M_k$, and at every terminal set the other two incident variables to one. The only remaining conditions are the ten vertex equations

$$\bigoplus_{i:e_i \ni v} q_i = 0. \tag{11}$$

Ordinary row reduction over $\mathbb{F}_2$ gives the following complete reduced systems. The set U lists the variables not fixed before reduction.

k	U	reduced equations
1	$4, 6, 9, 10, 14$	$(q_4, q_6, q_9, q_{10}, q_{14}) = (0, 0, 1, 0, 1)$
2	$1, 2, 3, 4, 5, 7$	$q_1 + q_5 = 1,\ q_2 + q_5 = 0,\ q_3 = 0,\ q_4 + q_5 = 0,\ q_7 = 0$
3	$0, 2, 10, 12, 14$	$(q_0, q_2, q_{10}, q_{12}, q_{14}) = (1, 0, 0, 0, 0)$
4	$0, 1, 8, 9, 11$	$0 = 1$
5	$3, 11, 12, 13, 14$	$0 = 1$
6	5	$0 = 1$
7	$6, 7, 8, 13, 14$	$0 = 1$

(12)

For an additional rank check, the coefficient/augmented ranks in the last four rows are respectively $4/5, 4/5, 1/2, 4/5$. Thus values 4 through 7 admit no forced-edge Eulerian set. Values 1 and 3 have one solution each, and value 2 has two solutions according to the free choice of q_5. Their circuit components are exactly

k	circuit edge sets	terminal counts
1	$\{0,1,9\}$; $\{3,5,7,8,12,13,14\}$	$1,3$
2	$\{0,1,9\}$; $\{6,8,11,12,14\}$	$1,3$
2	$\{0,2,4,5,9\}$; $\{6,8,11,12,14\}$	$1,3$
3	$\{0,1,9\}$; $\{3,5,6,7,11,13\}$	$1,3$

(13)

Equations (11) make the completeness of (12) a direct five-variable calculation in the largest case; substitution into (8) checks every listed circuit. Every possible circuit decomposition has odd terminal counts on both components. Lemma 3.2 therefore excludes two edge-disjoint T-joins for all seven M_k.

Finally, the colour column of (8), in edge order, is

$$(0,1,2,2,0,1,1,0,2,2,1,1,0,0,2). \tag{14}$$

At every vertex the three incident colours are $0,1,2$. For each colour, take the union of the other two colour classes. These three Eulerian edge sets cover every edge exactly twice; append two empty sets. Thus G has a standard 5-CDC. □

The frozen artifact contains a second checker which, without using Lemma 3.2, enumerates all T-joins in each complement. It finds $16, 16, 16, 16, 16, 8, 16$ joins for values $1, \ldots, 7$ and minimum pairwise intersection one in every case. This is a computational cross-check of the human proof, not a replacement for it.

6 The connected countermodel

We now refute (D). The proof is a composition argument from the ten-vertex block B of Theorem 5.1 and a second ten-vertex graph with three visibly non-cyclable monochromatic edges. No SAT certificate is needed for the main construction.

Let (G_1, f_1) and (G_2, f_2) be vertex-disjoint cubic flow graphs, and let

$$e_1 = x_1y_1 \in E(G_1), \qquad e_2 = x_2y_2 \in E(G_2)$$

have the same value b. Their *aligned two-sum* deletes e_1, e_2, adds x_1x_2, y_1y_2, and gives both new edges value b. The other pairing of the four ends is equivalent for the arguments below. The two new edges are called the connectors.

Lemma 6.1 (flow, structure, and circuit projection)**.** *An aligned two-sum of connected bridgeless cubic flow graphs is again a connected bridgeless cubic flow graph. It is simple when the summands are simple. Every circuit of the sum uses zero or both connectors. If it uses both, replacing its path through either summand by that summand's deleted edge gives a circuit in the graph before the sum. If the summands have standard 5-CDCs, then so does the sum.*

Proof. Every affected vertex loses and gains one incident edge of the same flow value b; hence conservation and degree three are preserved. Deleting a non-bridge from either connected summand leaves it connected, and the two connectors join the two remaining graphs, so the sum is connected.

Each connector lies on a circuit formed from the two connectors and paths between the deleted-edge ends in the two summands. An old edge lies on an old circuit; if that circuit used a deleted edge, replace the deleted edge by the corresponding path through the other summand. Thus every edge of the sum lies on a circuit, proving bridgelessness. Simplicity is immediate because the connectors join disjoint vertex sets.

A circuit meets the displayed two-edge cut evenly, so it uses zero or both connectors. In the latter case, deleting the two connectors cuts the circuit into one path in each summand, between the ends of the deleted edge. Restoring that edge closes either path to a circuit. This is the claimed projection.

For the cover assertion, use Proposition 2.1. Permute the five coordinates of one summand so the two deleted edges have the same two-subset label, and give both connectors that label. At each affected vertex one copy of that label is removed and one is added. Exact-two coverage and all five incidence parities are therefore preserved. □

Lemma 6.2 (non-cyclability is inherited by port replacement)**.** *Let $P \subseteq E(H)$ contain at least two edges, and suppose no circuit of H contains all of P. Replace every $p \in P$ by an aligned two-sum with a separate graph. Every circuit of the resulting graph is disjoint from one whole attached graph and its two connectors.*

Proof. If the circuit lies wholly in one attached graph, any other attached graph is missed. Otherwise project every traversed attachment back to its replaced edge using Lemma 6.1. The result is a circuit of H, so it omits some $p \in P$. The corresponding attached graph and both connectors were not traversed. □

Lemma 6.3 (one unchanged bad block blocks every value)**.** *Attach a copy of B by an aligned two-sum at an edge xy of value b. Suppose a later modification leaves the flow on the B-side and on both connectors unchanged. Then no value $k \neq 0$ is successful in the resulting global flow.*

Proof. Suppose instead that J_1, J_2 are edge-disjoint global ∂M_k-joins in the complement of M_k.

If $k = b$, both connectors are removed with M_k. Restricting J_1, J_2 to the isolated B-side gives two edge-disjoint $\partial M_k(B)$-joins in $B - M_k(B)$, contradicting Theorem 5.1.

Let $k \neq b$. Both connectors remain. For either J_i, the parity sum over all vertices on the B-side shows that it uses an even number of connectors: the terminals in that shore are exactly $\partial M_k(B)$, a set of even cardinality. Hence it uses zero or two connectors. If it uses none, its restriction is a $\partial M_k(B)$-join. If it uses both, delete the connectors and restore the port edge xy; the resulting edge set is again a $\partial M_k(B)$-join. Because J_1, J_2 are edge-disjoint, at most one of them uses both connectors, so at most one projection uses xy. The two projected joins are therefore edge-disjoint, again contradicting the defining property of B. □

6.1 A ten-vertex non-cyclable base

Let H have the following edge table. The column $c(e)$ is a proper three-edge-colouring, and the flow value is $1, 2, 3$ on colours $0, 1, 2$, respectively.

i	e_i	$c(e_i)$	$f(e_i)$	i	e_i	$c(e_i)$	$f(e_i)$
0	01	0	1	8	67	2	3
1	02	1	2	9	18	2	3
2	03	2	3	10	48	1	2
3	23	0	1	11	58	0	1
4	45	2	3	12	29	2	3
5	46	0	1	13	39	1	2
6	56	1	2	14	79	0	1
7	17	1	2				

(15)

Its graph6 record is `It?GYDKKO`. The table gives every vertex one edge of each colour, so H is cubic and the three incident flow values xor to zero. It is connected: from 0 one reaches $1, 2, 3$, then $7, 8, 9, 6$, and then $4, 5$. Every edge lies in the two-colour 2-factor obtained by omitting the third colour, so H is bridgeless.

Put

$$P = \{1, 6, 7\} = \{02, 56, 17\}. \tag{16}$$

All three edges have value 2, but no connected circuit contains all three. Indeed, suppose a circuit C contains P. It has edges outside the triangle on $\{0, 2, 3\}$, so it crosses that triangle's three-edge boundary in two edges. If $01 \notin C$, then $29, 39 \in C$; together with 02, the forced degrees at $0, 2, 3, 9$ make $0\,2\,9\,3\,0$ the whole circuit, a contradiction. Hence $01 \in C$.

Apply the same argument to the triangle on $\{4, 5, 6\}$. If $67 \notin C$, the forced edges make $8\,4\,6\,5\,8$ the whole circuit. Thus $67 \in C$, and exactly one of $48, 58$ lies in C. At vertex 7, the edges $17, 67$ exclude 79. At vertex 1, the edges $17, 01$ exclude 18. Vertex 8 is consequently incident with exactly one edge of C, namely one of $48, 58$, which is impossible for a circuit. This proves the non-cyclability of P without computation.

Theorem 6.4 (40-vertex countermodel to one-circuit domination)**.** *There is an explicit simple connected bridgeless cubic graph G on 40 vertices and 60 edges with a nowhere-zero $\mathbb{F}_2^3$-flow f such that $f \notin \mathcal{G}(G)$ and every allowed switch of f on one connected circuit also lies outside $\mathcal{G}(G)$. Moreover, G has an explicit three-cycle double cover. Thus* (D) *is false, but G is not a counterexample to the five-cycle double cover conjecture.*

Proof. For each of the three edges $p \in P$, replace p by an aligned two-sum with a fresh copy of B, using edge 10 of B as the port. Both edges have value 2. Repeated use of Lemma 6.1 proves that the resulting G, f is simple, connected, bridgeless, cubic, and nowhere zero. The counts are

$$|V(G)| = 10 + 3 \cdot 10 = 40, \qquad |E(G)| = 15 + 3 \cdot 15 = 60.$$

By Lemma 6.2, every connected circuit C of G misses one whole copy of B and its connectors. An allowed switch on C leaves that block unchanged, so Lemma 6.3 shows that all seven values of the switched flow remain unsuccessful. The original flow is bad by the same lemma, using any attached block.

For the positive assertion, edge 10 of B and all three edges of P have colour 1 in the displayed proper three-edge-colourings. Give both connectors colour 1. At each affected vertex an edge of colour 1 is replaced by another, so the final graph remains properly three-edge-coloured. For each colour, take the union of the other two colour classes. These three Eulerian edge sets form a cycle double cover, each with 40 edges. Appending two empty sets gives a standard 5-CDC. □

Graph metadata. The nauty-canonical graph6 encoding is obtained by concatenating the following two displayed lines:

```
gt?GG?@?GB?@?D??@O?K???_?CGA_??A???@??OO??C??@_??C???G?_???G???D???C??
??_G???G????OC????B????K?????G????@@O????KO????BA?GC??@O??O??.
```

NetworkX and nauty `planarg` report that G is planar, and its vertex- and edge-connectivity are both two. These are metadata about the route countermodel, not evidence against five-CDC; the explicit 3-CDC above proves the opposite for this graph.

The construction has a visible nontrivial two-edge cut at each block attachment, so it does not address (D) restricted to cyclically 4-edge-connected graphs.

The complete construction, flow, three-cover labels, checker, and frozen replay are in

`search/connected-one-switch-countermodel-40v-20260726/`.

The checker does not import the producer. It reconstructs the three two-sums; verifies simplicity, cubicity, connectedness, every single-edge deletion, and every flow equation; enumerates all 30 connected circuits of H and all 6,780 connected circuits of G; confirms that every expanded circuit misses an entire bad block; re-enumerates all local T-joins; and checks the three-cover coordinate sizes

$$40, \quad 40, \quad 40, \quad 0, \quad 0.$$

This is an AI-assisted mechanical replay, not independent human review.

7 A potential-compatible pure-merge countermodel

We now study a different, narrower route from a fixed $\mathbb{F}_2^3$-flow to five Eulerian coordinates. The potential construction below is the flow-to-pair-label construction in the July 2026 cycle-double-cover proof, as presented by Oum and Kintali [9, Section 5] and [6]. We use *potential-compatible* only as local shorthand for covers arising from these equations; it is not terminology attributed to those authors.

7.1 Potentials, pair labels, and pure merges

Let G be cubic and let $\phi : E(G) \to \mathbb{F}_2^3 \setminus \{0\}$ be a flow. A family $(t_v : v \in V(G))$ of vertex potentials is *compatible* when, for every edge $e = uv$,

$$t_u + t_v \in d_e + \langle \phi(e) \rangle, \qquad d_e = \phi(f_u) + \phi(f_v), \tag{P}$$

where $f_u \neq e$ and $f_v \neq e$ are incident with u and v. The choice of either other edge at an endpoint is immaterial because the three incident flow values sum to zero. A compatible potential defines the two-element label

$$P_e = t_u + \phi(f_u) + \langle \phi(e) \rangle = t_v + \phi(f_v) + \langle \phi(e) \rangle. \tag{L}$$

For $s \in \mathbb{F}_2^3$, let $A_s = \{e : s \in P_e\}$. If e, f, g are the three edges at v, then their labels at that end are

$$\{t_v + \phi(f), t_v + \phi(g)\}, \quad \{t_v + \phi(e), t_v + \phi(g)\}, \quad \{t_v + \phi(e), t_v + \phi(f)\}.$$

Every coordinate consequently occurs zero or twice at v. Thus all eight A_s are Eulerian and cover each edge exactly twice. We call such a family a *potential-compatible eight-cover for the fixed flow* ϕ.

A *pure merge* assigns the eight coordinates to at most five classes and takes the symmetric difference of the A_s inside each class. It changes no edge within a coordinate before this final merging step. The *coordinate co-occurrence graph* $Q(P)$ has vertex set $\mathbb{F}_2^3$, with ab an edge when $P_e = \{a, b\}$ for some $e \in E(G)$.

Proposition 7.1 (pure-merge colouring equivalence)**.** *A potential-compatible eight-cover admits a pure merge into at most five Eulerian edge-subsets, still covering every edge exactly twice, if and only if $Q(P)$ is properly five-colourable.*

Proof. Colour each coordinate by its merge class. An edge with label $\{a, b\}$ survives in two output coordinates exactly when a and b have different colours; if their colours agree, its two occurrences cancel in the same symmetric difference. Hence a valid merge gives a proper colouring of $Q(P)$. Conversely, use the colour classes of a proper colouring as merge classes. Symmetric differences of Eulerian edge sets are Eulerian, and every original edge survives once in each of two different classes. □

This equivalence concerns only a pure identification of the eight coordinates. It says nothing against a five-cycle double cover obtained by a different construction.

7.2 A potential-rigid twelve-vertex cap

Let R be the cubic graph with graph6 record `K??FEaKR@oE_`. The following table freezes its edge order, flow, proper edge-colouring c, the vector d_e from (P), and one compatible pair label from (L).

i	e_i	$\phi(e_i)$	$c(e_i)$	d_i	P_i	i	e_i	$\phi(e_i)$	$c(e_i)$	d_i	P_i
0	06	1	0	6	23	9	37	5	2	5	36
1	07	2	1	6	13	10	3, 10	7	1	6	34
2	08	3	2	4	12	11	3, 11	2	0	3	46
3	16	4	1	6	26	12	48	5	0	2	14
4	17	7	0	6	16	13	49	1	1	6	01
5	19	3	2	5	12	14	4, 10	4	2	6	04
6	26	5	2	2	36	15	58	6	1	1	24
7	2, 10	3	0	2	03	16	59	2	0	5	02
8	2, 11	6	1	7	06	17	5, 11	4	2	0	04

(18)

Here pairs such as 23 mean $\{2, 3\}$, not an integer. The three incident flow values xor to zero at every vertex, and the colour column is proper. In particular, every edge of R lies in a two-colour 2-factor, so R is bridgeless; connectedness is immediate from the displayed incidence table.

Use edge $1 = 07$ as the port and delete it. We next give a direct row-reduction proof that the retained compatibility equations determine all twelve potentials up to adding one common vector.

Fix $t_0 = 0$. On the spanning tree whose edges, in order, are

$$0,\ 2,\ 3,\ 6,\ 12,\ 15,\ 4,\ 5,\ 7,\ 8,\ 9,$$

write $t_u + t_v = d_e + \lambda_j \phi(e)$ for the j-th displayed tree edge $e = uv$. These tree equations express all t_v in the eleven bits $\lambda_0, \ldots, \lambda_{10}$. Substitution in the six remaining internal edges $10, 11, 13, 14, 16, 17,$

followed by elementary xor row operations, gives

$$\begin{array}{rcl@{\qquad}rcl}
\lambda_3+\lambda_{10} &=& 0 & \lambda_2+\lambda_6+\lambda_9 &=& 0\\
\lambda_2+\lambda_8 &=& 1 & \lambda_1+\lambda_7 &=& 0\\
\lambda_6 &=& 1 & \lambda_2+\lambda_5 &=& 1\\
\lambda_2+\lambda_4 &=& 1 & \lambda_0+\lambda_1+\lambda_3 &=& 0\\
\lambda_1+\lambda_2 &=& 0 & \lambda_1 &=& 1\\
\lambda_0 &=& 1. & & &
\end{array}$$

The unique solution and its back-substitution are

$$\begin{aligned}(\lambda_0,\ldots,\lambda_{10}) &= (1,1,1,0,0,0,1,1,0,0,0),\\ (t_0,\ldots,t_{11}) &= (0,5,5,1,5,6,7,4,7,3,7,2).\end{aligned} \tag{19}$$

Thus the 37 scalar equations—two for each retained edge and three fixing t_0—have rank 36.

Computing (L) from (19), the internal labels and the dangling port label 13, calculated from either end of the deleted port, use exactly

$$\begin{array}{c}01,\ 02,\ 03,\ 04,\ 06,\qquad 12,\ 13,\ 14,\ 16,\\ 23,\ 24,\ 26,\qquad 34,\ 36,\qquad 46.\end{array} \tag{20}$$

These are all 15 pairs on $\{0,1,2,3,4,6\}$, so the co-occurrence graph contains K_6.

Lemma 7.2 (rigid-cap obstruction)**.** *Suppose a copy of $R-e_1$, with the displayed internal flow and two dangling flow-2 connectors, occurs in a larger cubic flow graph. Every compatible global potential forces a coordinate K_6 on the cap and its connectors. Hence no compatible eight-cover for that fixed global flow admits a pure merge into at most five coordinates.*

Proof. Restrict a compatible global potential to the cap. The retained equations just solved show that it is (19) plus one common vector $a \in \mathbb{F}_2^3$. Formula (L) at the cap ends gives the translated dangling label $13 + a$ on the connectors. Thus (20) is translated by a, and still induces a K_6. Proposition 7.1 completes the proof. □

7.3 The 46-vertex one-switch countermodel

Theorem 7.3 (46-vertex pure-merge countermodel)**.** *There is an explicit simple connected bridgeless cubic graph $G^\star$ on 46 vertices and 69 edges, with a nowhere-zero $\mathbb{F}_2^3$-flow ϕ, such that:*

1. *no potential-compatible eight-cover for ϕ admits a pure merge into at most five Eulerian edge-subsets; and*
2. *for every connected circuit C and every nonzero $a \in \mathbb{F}_2^3$ for which $\phi' = \phi + a\chi_C$ remains nowhere zero, no potential-compatible eight-cover for ϕ' admits such a pure merge.*

Moreover, $G^\star$ is Tait-colourable and has an explicit three-cycle double cover. It is therefore not a counterexample to the five-cycle double cover conjecture.

Proof. Start from the base H in (15). For each $p \in P = \{02, 56, 17\}$, replace p by an aligned two-sum with a fresh copy of R, using edge $1 = 07$ as the cap port. Both the base ports and the cap port have flow value 2. The resulting counts are

$$|V(G^\star)| = 10 + 3\cdot 12 = 46, \qquad |E(G^\star)| = 15 + 3\cdot 18 = 69.$$

Lemma 6.1, together with the incidence table above, proves that the displayed graph and flow are simple, connected, bridgeless, cubic, and nowhere zero.

The original flow has a compatible potential: put $t_v = 0$ on the base and use (19) on each cap. The base pair labels are $23, 13, 12$ on colours $0, 1, 2$, respectively, so both sides assign 13 to every connector. Lemma 7.2 applied to any cap proves (i).

Let C be a connected circuit of $G^\star$. The non-cyclability proof following (16), equivalently Lemma 6.2, shows that C leaves one whole cap and both its connectors untouched. For every allowed switch $\phi' = \phi + a\chi_C$, the flow on that cap and its connectors is unchanged. Restricting any potential compatible with ϕ' to the untouched cap and applying Lemma 7.2 gives a forced K_6. This proves (ii).

Finally, the base ports and cap port all have colour 1 in the proper edge-colourings in (15) and (18). Give the connectors colour 1. This produces a proper three-edge-colouring of $G^\star$. If its colour classes are M_0, M_1, M_2, then

$$E(G^\star) \setminus M_0, \quad E(G^\star) \setminus M_1, \quad E(G^\star) \setminus M_2$$

are Eulerian and cover every edge exactly twice. Each has 46 edges; append two empty sets for a standard 5-CDC with sizes $46, 46, 46, 0, 0$. ☐

The construction has three visible nontrivial two-edge cuts, one at each cap. It therefore does not refute a pure-merge statement restricted to cyclically 4-edge-connected cubic graphs. The frozen construction metadata reports that $G^\star$ is nonplanar; this fact is not used in the proof, and the independent checker does not recheck planarity.

Exact encoding and check. The nauty-canonical graph6 encoding is obtained by concatenating:

```
ms????@?G@???P?D?@_?o?_?A?????O?CC?I@?@_O?W??Q???W??_?O????G???K????G???
?OA????W????????E????I?????A_????S????B????????????????????C?????WO????@
Q?????@`?????@__????B?O????B??.
```

The complete construction and SAT-free proof package are in

`search/fano-pure-merge-one-switch-countermodel-46v-20260726/`.

Its independently written standard-library checker reconstructs the graph from the ten- and twelve-vertex constants; checks all 69 single-edge deletions, every flow and parity equation, and the explicit three-cover; independently row-reduces the cap system to rank 36 and checks the forced K_6; and enumerates all 30 base circuits. As a further cross-check, it enumerates all 128 globally compatible gauged potentials for the original flow: every co-occurrence graph contains a K_6, and none is five-colourable. The checker and this audit remain AI-assisted rather than independent human verification.

8 Finite reconfiguration frontier

These searches preceded the construction of Theorem 6.4. They are retained because they show how far the false statement survives finite testing, not because they support it as a universal claim. We state only what the artifacts certify. They test the successful-value condition and do not test the distinct pure-merge property in Theorem 7.3. A *hard graph* in this section is connected, simple, bridgeless, cubic, and not 3-edge-colourable. A *strict snark* is additionally cyclically 4-edge-connected and has girth at least five. The flow enumeration quotients by $\mathrm{GL}(3, 2)$. A bad flow uses all seven nonzero values, since an empty value class has $M_k = \varnothing$ and is immediately successful.

The retained normalization first fixes the ordered value line at a root vertex and then removes its residual stabilizer.

Computational Observation. *The retained programs report the following exact finite totals.*

scope	flow orbits	bad	reported repair	unrepaired
all bridgeless cubic, orders 4–10	3,295	5	5	0
all bridgeless cubic, order 12	73,152	304	304	0
all bridgeless cubic, order 14	2,216,590	18,152	18,152	0
26 hard graphs, order 16	600,440	10,413	10,413	0
179 hard graphs, order 18	21,324,510	757,221	757,221	0
1,388 hard graphs, order 20	847,539,220	49,752,091	49,752,091	0
20 strict snarks, order 22	63,251,330	279,836	279,836	0
38 retained strict snarks, order 24	617,079,260	5,161,169	5,161,169	0
one retained girth-6 snark, order 28	432,613,280	3,400,482	3,400,482	0

(17)

Here the first five retained producers test arbitrary nonempty binary cycles, which may have more than one circuit component. Such a switch decomposes into a valid sequence of circuit switches because the switched cycle avoids the a-class, which remains fixed throughout. But a one-binary-cycle repair is not by itself a certificate of *one-circuit adjacency* from the starting flow.

The complete order-20 hard-graph computation was repeated with the switch restricted to a connected circuit. The elementary-only C++ ledger again contains all 847,539,220 orbit representatives and reports all 49,752,091 bad representatives repaired in one circuit, with zero unrepaired. The same elementary-circuit enumerator covers the complete frozen sets of 20 strict order-22 snarks and 38 strict order-24 snarks. At order 22, among 63,251,330 flow orbits, all 279,836 initially bad representatives are repaired. At order 24, among 617,079,260 flow orbits, 611,918,091 are initially good and all 5,161,169 initially bad representatives are repaired. Neither scope contains a switch-local bad flow. The order-24 source is the complete output of one retained Snarkhunter 2.0b command and is frozen with the ledger. The verifier checks all 38 distinct graph identities and their strict-snark premises, but does not independently regenerate the canonical list; census completeness therefore relies on Snarkhunter. These last three rows are direct finite tests of the now-refuted statement (D), with no extrapolation beyond the displayed scopes. No analogous full elementary-only ledger is claimed here for the earlier rows. The 40-vertex countermodel demonstrates that these zero-obstruction totals are a finite phenomenon.

Separately, the enumerator exhausts 432,613,280 flow orbits on one retained cyclically 4-edge-connected girth-6 snark of order 28. All 3,400,482 initially bad representatives are repaired in one circuit. This is one graph, not a complete order-28 graph census.

Artifacts and checks

The ten-vertex block package is

`search/fano-value-class-flow-countermodel-20260726/`,

the connected countermodel package is

`search/connected-one-switch-countermodel-40v-20260726/`,

the pure-merge countermodel package is

search/fano-pure-merge-one-switch-countermodel-46v-20260726/,

and the finite-frontier package is

search/fano-flow-one-switch-frontier-20260726/.

From the repository root, the principal checks are

```
(cd search/fano-value-class-flow-countermodel-20260726 &&
 python3 independent_checker.py &&
 shasum -a 256 -c SHA256SUMS)

(cd search/fano-flow-one-switch-frontier-20260726 &&
 python3 verify.py &&
 shasum -a 256 -c SHA256SUMS)

(cd search/connected-one-switch-countermodel-40v-20260726 &&
 python3 independent_checker.py &&
 shasum -a 256 -c SHA256SUMS)

(cd search/fano-pure-merge-one-switch-countermodel-46v-20260726 &&
 python3 independent_checker.py)
```

On the retained environment the first three package manifests pass, and the fourth command reproduces the frozen 46-vertex checker output. The block checker reconstructs graph6 identity, cubicity, connectedness, bridgelessness by every single-edge deletion, flow conservation, all seven matching supports, literal T-join enumeration, the colouring, and the resulting five-cover.

For the frontier, a separately written C++ implementation matches the complete Python order-14 totals. It is also the producer for the two order-20 ledgers and the connected-circuit order-22, order-24, and one-graph order-28 ledgers. The Python verifier checks the frozen aggregate identities and semantically reconstructs the first retained bad-flow repair on 432 graph rows. It does not re-enumerate every order-16, order-18, order-20, order-22, or order-24 flow orbit, nor the retained order-28 graph.

Limitations

1. The finite frontier does not contradict Theorem 6.4; its tested graph orders and scopes simply do not include the constructed obstruction. It concerns a different property from Theorem 7.3.

2. At orders 16, 18, and 20, the graph scope contains only hard graphs, and at orders 22 and 24 only strict snarks, not every bridgeless cubic graph. This matters because Theorems 6.4 and 7.3 are both 3-edge-colourable. The order-24 census additionally relies on the retained Snarkhunter run rather than an independent second generation. The order-28 row is explicitly only one graph.

3. The programs and their cross-checks were all AI-assisted. They are implementation diversity, not independent human validation.

4. The full positive model or repair witness for every one of the hundreds of millions of orbit decisions is not retained as a separately checkable certificate archive.

5. The computation uses simple graphs. The structural statements above allow loopless cubic multigraphs, but no finite conclusion is extrapolated to untested parallel-edge instances.

6. No orientability condition is encoded or inferred.

9 What remains open

Theorem 6.4 ends the proposed one-connected-circuit domination route, and Theorem 7.3 ends the proposed route of applying one circuit switch and then purely merging the eight coordinates supplied by potential-compatible labels. Neither theorem produces a graph without a 5-CDC: both graphs have displayed 3-CDCs. The general five-cycle double cover conjecture therefore remains exactly as open as before this work. In particular, the second theorem does not exclude changing coordinates by operations more general than a pure merge, choosing a different flow after several switches, or constructing a 5-CDC without the potential eight-cover.

Both constructions have nontrivial two-edge cuts. Hence neither theorem settles the corresponding one-circuit assertion for cyclically 4-edge-connected cubic graphs. No reduction from either restricted assertion to the general 5-CDC conjecture is claimed here.

The structural statements in Sections 2 and 3 may still be useful independently. A successful value is a sufficient condition for a cover, and the four canonical ∂M_k-joins retain their tetrahedral intersection pattern. What fails is the claim that one circuit switch must reach such a value. Any future proof strategy must either use longer/global reconfiguration, combine information from unsuccessful values without requiring one to become successful, or leave this flow framework. For the pure-merge route, a future argument would have to permit a richer transformation than merely identifying coordinates, use more than one local flow switch, or establish a restricted theorem on a domain not covered by the cap construction.

AI-use statement

The page-one disclosure is part of the mathematical record of this draft and should remain in every derivative version. OpenAI Codex agents materially contributed to the definitions selected for study, conjecture, proof search, the finite witnesses, both two-sum compositions, the potential-rigid cap, source code, SAT encoding, checkers, experiments, literature search, and exposition. A second Codex agent performed an adversarial audit, but that is not independent human peer review. The recent flow-reconfiguration framework is attributed to [2, 1]; the matching/4-flow criterion is attributed to [4]; and the potential construction is attributed to [9, 6]. The two-sum proofs, the three-edge non-cyclability argument, and the rigid-cap row reduction are written for direct human checking, but no independent human check is asserted to have occurred. Any human authors should record their verification, independent reconstruction, and rerun history before considering public submission.

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